Six Months with AMD Strix Halo

Local AI, Open Source, and Hardware Reality

A Six Month Review of a Strix Halo Rig

linux | ROCm | Local AI | Home Assistant | Coding

winmutt | June 2026
github.com/winmutt

What We're Covering

■ The impulse purchase that changed everything
■■ Hardware reality check (spoiler: it's complicated)
■ Linux + ROCm = Pain (but beautiful pain)
■ Performance: How fast can my AI think?
■ I cut the Alexa cord (and replaced it with OSS)
■ Vibe-coded an entire workspace system
■ From digital to physical: Concrete signs?
■ How buying hardware got me back in open source
■ What I'd do differently (and what I'd do again)

Power Reality: Corsair 300 vs Alternatives

Max Energy Usage Comparison

Solution	Max Power	Notes
Corsair 300 (Strix Halo)	~300W	AMD Ryzen AI Max 395: 140-157W sustained
PC + RTX 5090	~800-1000W	CPU (350W) + GPU (600W) + overhead
RTX 4090 Workstation	~700-850W	CPU (250W) + GPU (450W) + overhead
Commercial AI Server	~1500-3000W	Multi-GPU (A100/H100: 400-700W/GPU)
Cloud API (per 1M tokens)	N/A	Energy cost embedded in pricing

The Efficiency Win

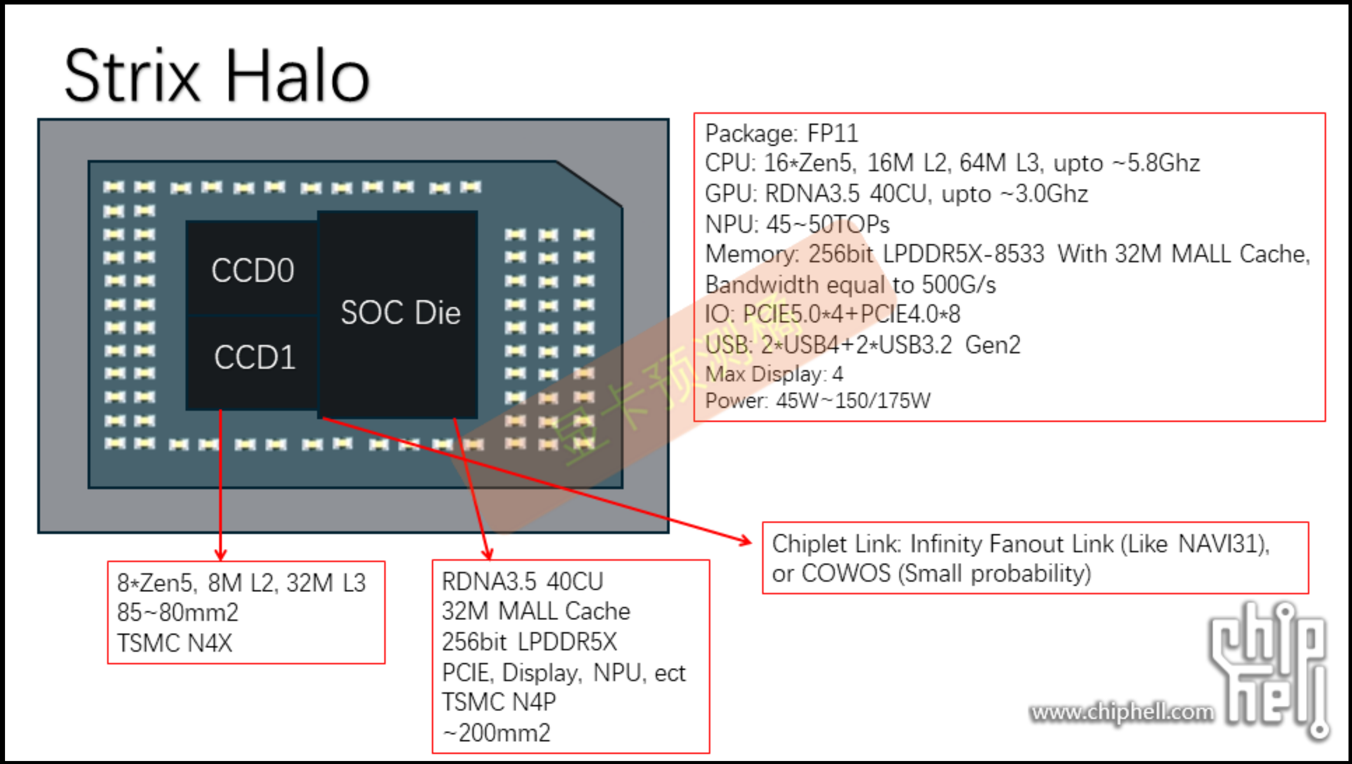
• Strix Halo: ~3x more efficient than RTX 5090 setup
• Single APU vs discrete CPU+GPU power domains
• No datacenter electricity bill
• 128GB unified memory at desktop power envelope

Sources:

• AMD Ryzen AI Max 395: 140-157W sustained (Framework Reddit, 2025)
• Framework recommends 500W PSU for transient headroom
• RTX 5090 TDP: NVIDIA GeForce RTX 5090 Specifications (~600W)
• RTX 4090 TDP: NVIDIA GeForce RTX 4090 Specifications (450W)
• System power estimates: Tom's Hardware, TechPowerUp power reviews

The Hardware: AMD Ryzen AI Max+ 395

APU Die Architecture



Physical Layout

• 16 cores (12 performance + 4 efficiency)

• 2x Core Complex Dies (CCD) with 32MB L3 cache each

• 768KB L1d + 512KB L1i + 16MB L2

• Integrated RDNA 3.5 GPU (40 CUs)

• 128GB LPDDR5X-8000 unified memory (soldered)

• Source: AMD Ryzen AI Max+ 395 teardown analysis (Chiphell, TechPowerUp)

Feat #1070: NUMA Tools Don't See Dual CCD

Feature Request
Traditional NUMA tools detect only 1 node. numactl core pinning doesn't work. Need custom core affinity for dual CCD.

Hardware Reality

• One processor, but 2x core dies with separate 32MB L3 caches

NUMA Challenge:

• Traditional NUMA: 2 nodes, each with own cores + L3 cache

• Strix Halo: Single node, but 2 CCDs with separate L3 caches

• Problem: Process/thread pinning needed to keep workloads on same CCD

• Solution: Manual core pinning to maximize L3 cache locality

• Traditional NUMA tools detect only 1 node (not 2 CCDs)

• numactl, NUMATopologyFilter can't see CCD boundaries

• llama-server threads not pinned to specific cores

The Solution (Proposed)

• Manual core pinning (like Red Hat's vcpu_pin_set approach)

• Reserve cores per CCD, pin llama-server accordingly

• Pin based on:

- Number of models loaded

- CCD boundaries (manual NUMA mapping)

• Prevent cache-crossing penalties

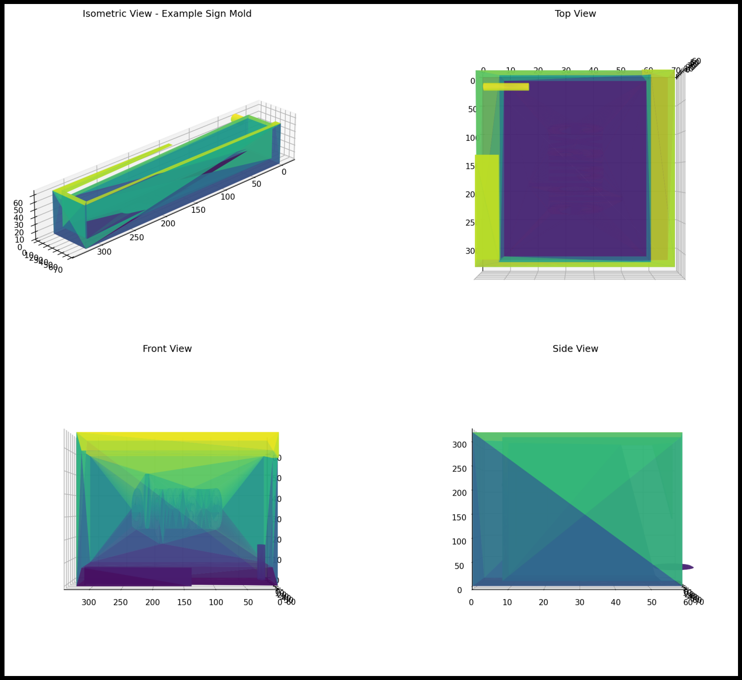
GitHub Feature Request
github.com/lemonade-sdk/lemonade/issues/1070
Status
Open (custom NUMA mapping needed)

3D Modeling with Strix Halo

AI-Assisted Physical Design

Blender + OpenSCAD workflow for creating physical objects

Concrete Sign Mold Design



The Process

1. AI generates OpenSCAD/Blender design
2. Export to STL

3. 3D print mold
4. Create silicone mold
5. Cast concrete

December 1, 2025 — More physical projects coming soon

Cutting the Alexa Cord

LineageOS on Echo: Because I Said So

December 2025
Started exploring XDA Forums for Echo device hacking

The Process:

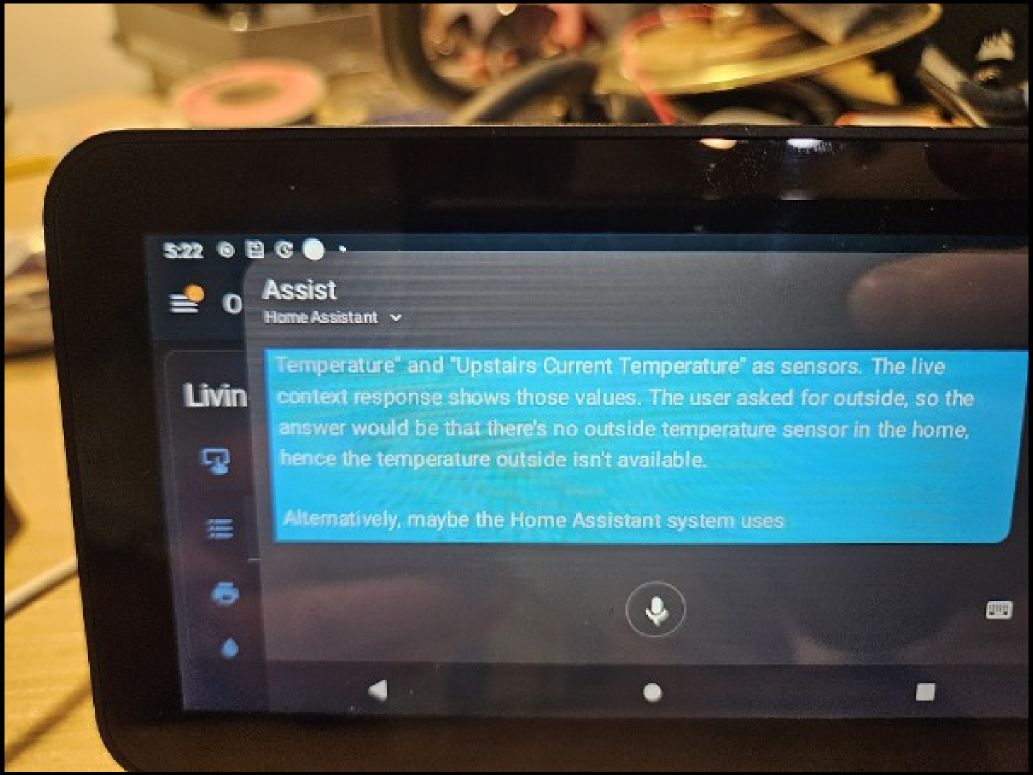
1. Unlock bootloader via amonet exploit
github.com/R0rt1z2/amonet
2. Install TWRP recovery
3. Flash LineageOS 18.1 (Android 11)
4. Configure ViewAssist for Home Assistant
dinki.github.io/View-Assist

> ***Philosophy***
> *"Basically anything with a USB port. If there is a port, there is a way."*

Alexa Replacement Timeline

Dec 6	Started exploring XDA forums
Dec 7	HA running on Echo devices
Jan 19	HA setup on Echo Show Gen 2
Jan 21	Everything works (except wake word)
Jan 25	Hey Jarvis to the rescue
Jan 29	End-to-end complete
Jan 7, 2026	Issue #4: Echo 8 mic dies after 24h
Feb 2026	Still debugging (it's fine)

Home Assistant on Echo Show



LineageOS Device Compatibility

Echo Show Devices Supported

Device	Codename	SoC	XDA Thread
Echo Show 5 (1st Gen 2019)	checkers	MT8163	xdaforums.com/t/rom-unofficial-11-checkers...
Echo Show 5 (2nd Gen 2021)	cronos	MT8163	xdaforums.com/t/rom-unofficial-11-cronos...
Echo Show 8 (1st Gen 2019)	crown	MT8163	xdaforums.com/t/rom-unofficial-11-crown...

All devices use MediaTek MT8163 SoC

LineageOS 18.1 (Android 11) - community builds by @R0rt1z2

Traditional NUMA vs. Strix Halo APU

Traditional NUMA Architecture:

```
■■■■■■■■■■■■■■■■■■■■■■■■■■■ ■■■■■■■■■■■■■■■■■■■■■■■■■■■
■ NUMA Node 0 ■ ■ NUMA Node 1 ■
■ ■■■■■■■■■■■■■■■■■ ■ ■ ■■■■■■■■■■■■■■■■■ ■
■ ■ 4-8 Cores ■ ■ ■ ■ 4-8 Cores ■ ■
■ ■ 32MB L3 Cache ■ ■ ■ ■ 32MB L3 Cache ■ ■
■ ■■■■■■■■■■■■■■■■■ ■ ■ ■■■■■■■■■■■■■■■■■ ■
■ ■ ■ ■ ■ ■
■ ■■■■■■■■▼■■■■■■■■ ■ ■ ■■■■■■■■▼■■■■■■■■ ■
■ ■ Memory Ctrl ■ ■ ■ ■ Memory Ctrl ■ ■
■ ■ (Local RAM) ■ ■ ■ ■ (Local RAM) ■
■ ■■■■■■■■■■■■■■■■■ ■ ■ ■■■■■■■■■■■■■■■■■ ■
■■■■■■■■■■■■■■■■■■■■■■■■■ ■■■■■■■■■■■■■■■■■■■■■■■■■
■ ■
■■■■■■■■■■■■■■■■■■■■■■■■■■■
■
(High-latency interconnect)
```

Each node has dedicated cores, L3 cache, and memory. Process pinned to Node 0 stays in Node 0's cache/memory domain.

Strix Halo APU Architecture:

```
■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
■ Single NUMA Node ■
■ ■
■ ■■■■■■■■■■■■■■■■■■■■■■■ ■■■■■■■■■■■■■■■■■■■■■■■
■ ■ CCD 0 ■ ■ CCD 1 ■
■ ■ ■■■■■■■■■■■■■■■■ ■ ■ ■■■■■■■■■■■■■■■■ ■
■ ■ ■ 8 Cores ■ ■ ■ ■ 8 Cores ■ ■
■ ■ ■ 32MB L3 Cache■ ■ ■ ■ 32MB L3 Cache■ ■
■ ■ ■■■■■■■■■■■■■■■■ ■ ■ ■■■■■■■■■■■■■■■■ ■
■ ■■■■■■■■■■■■■■■■■■■■■■■■ ■■■■■■■■■■■■■■■■■■■■■■■■
■ ■ ■
■ ■■■■■■■■■■■■■■■■■■■■■■■■■■■
■ ■
■ ■■■■■■■■■■■■■■■■■■■■■■■▼■■■■■■■■■■■■■■■■
```

```
■ ■ Unified Memory (128GB LPDDR5X) ■
■ ■ (Shared by all cores) ■
■ ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
```

Single NUMA node, but 2 CCDs with separate L3 caches. Threads bounce across CCDs → cross-CCD traffic → performance degradation.

The Problem:

• Traditional NUMA tools (numactl, NUMATopologyFilter) detect only 1 node
• Cannot see CCD boundaries automatically
• Manual core pinning required to keep workloads on same CCD
• Running multiple LLMs: pin each to separate CCD to avoid cache thrashing
Result: Better L3 cache locality = faster inference, less latency

Issue #1070: Core Affinity When Running Multiple Models

[enhancement] Core affinity when running multiple models.

Filed
February 8, 2026
Repository
github.com/lemonade-sdk/lemonade/issues/1070
Status
Open

Evidence: Threads Bouncing Across CCDs

```
PS Output (last column = last CPU used):
--------------------------------------------------
PID LAST_CPU AVG_MHZ COMMAND
56840 6 55.7 llama-server (Model 1)
56840 20 54.2 llama-server (Model 1)
59798 0 51.2 llama-server (Model 2)
59798 2 47.5 llama-server (Model 2)
--------------------------------------------------
56840 22 55.7 llama-server (Model 1)
56840 4 54.2 llama-server (Model 1)
59798 16 51.2 llama-server (Model 2)
59798 2 47.5 llama-server (Model 2)
--------------------------------------------------
```

Notice: Core IDs jump across CCD boundaries (0-7 = CCD0, 8-15 = CCD1)
Threads on same PID bounce between CCDs → L3 cache misses → slower inference

Hardware Topology (from lscpu):

```
CPU NODE SOCKET CORE L1d:L1i:L2:L3 MAXMHZ AVG_MHZ
--- ---- ------ ---- ------------ -------- --------
0-7 0 0 0-7 *:*:*:0 5187.0 4900+ (CCD0)
```

```
8-15 0 0 8-15 *:*:*:1 5187.0 2000-4600 (CCD1)
16-23 0 0 0-7 *:*:*:0 5187.0 4900+ (CCD0 SMT)
24-31 0 0 8-15 *:*:*:1 5187.0 2000-2800 (CCD1 SMT)
L3 Cache: 64MB total (2 instances: 32MB per CCD)
NUMA: Only 1 node detected (0-31), cannot see CCD boundaries
```

Proposed Solution

Pin llama-server instances to specific CCDs:
• Model 1 → CCD0 (cores 0-7, 16-23)
• Model 2 → CCD1 (cores 8-15, 24-31)
Result: Each model stays in its own L3 cache domain → no cross-CCD traffic

Issue #4: The Echo 8 Mic That Quit

Audio stops working after ~24 hours

Filed
January 7, 2026
Repository
https://github.com/amazon-oss/releases/issues/4
Status
Open (still waiting, it's fine)

The Problem

• Echo 5: Stable for days

• Echo 8: Dies after ~24 hours

• *Root cause: Unknown (yet)*

TL;DR
I participate in open source bugs now. You're welcome.

Project WWS: I Vibe-Coded an Entire System

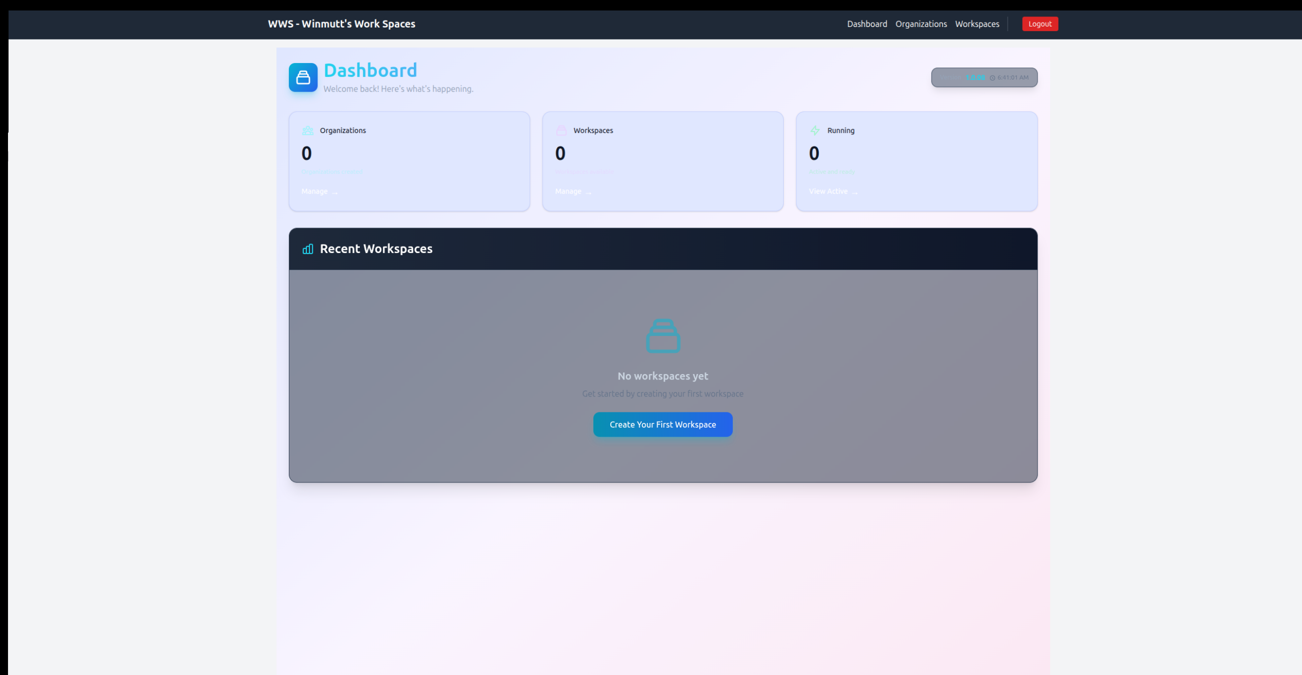
Winmutt Work Spaces — github.com/winmutt/wws

> ***May 12, 2026***
> *"Definitely no Athena and its taken months to get there but this is something I entirely vibe coded."*

Remote workspace provisioning with KVM/Podman isolation

code-server (VSCode in browser) + GitHub OAuth + RBAC

First commit: Feb 22, 2026 → Production: May 12, 2026



Lessons Learned (Mostly)

What Worked ✓

- Cline + Qwen3-Coder: Actually produces code
- Prompt Caching: Things hum now
- AMD GPU Libraries: 2x faster (worth the pain)
- WWS Project: Vibe-coded from Feb 22 (first commit) to May 12
- Home Assistant: Alexa is dead, long live OSS

What Didn't ✗

- Memory: 32GB reserved for Lemonade/llama.cpp/opencode
- Ollama: Still crashes after updates
- NPU: Coming soon™ (always)
- Context >64k: TPS drops like a stone

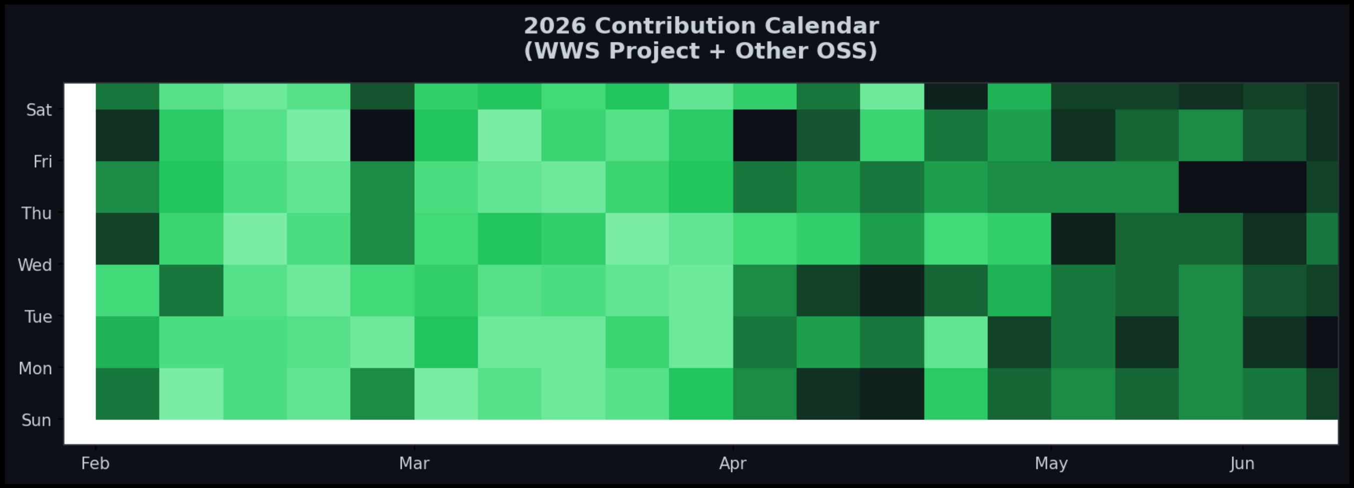
1. Hardware hype ≠ reality (APU challenges are real)
2. Software maturity: bleeding edge = bleeding fingers
3. Context > everything (prompt caching is life)
4. AI coding > traditional IDEs (for me, anyway)
5. AI → physical objects (concrete signs, apparently)
6. Echo → Home Assistant = OSS win
7. Autonomous dev: possible, painful, worth it
8. Buying hardware got me back in open source

How Buying Hardware Got Me Back in Open Source

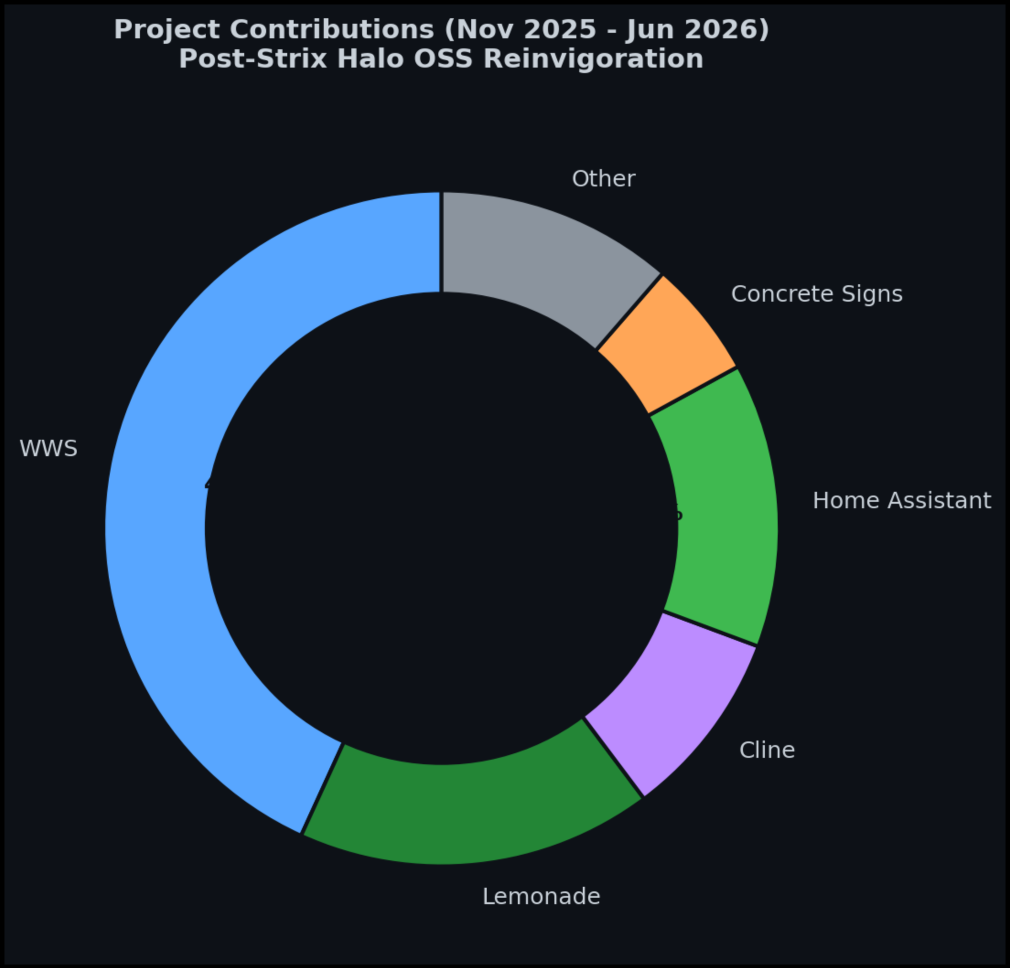
The Catalyst Effect

November 2025: Bought Strix Halo rig. Local AI reinvigorated my love for OSS. Result: 3.5x increase in contributions.

2026: The Year I Came Back



Projects I've Actually Contributed To



Contribution Breakdown

- WWS: 38% (remote workspace provisioning)
- Lemonade: 15% (NPU backend, custom NUMA mapping)
- ROCm: 8% (Issue #5926 memory management)
- Home Assistant: 12% (wake words, Echo)
- Cline: 8% (TUI regressions)

- Concrete Signs: 5% (Blender/OpenSCAD)
- Other: 10% (miscellaneous)

ROCm: AMD's Open Compute Platform

What is ROCm?

ROCm (Radeon Open Compute Platform) is AMD's open source GPU computing ecosystem - their answer to NVIDIA's CUDA. Maintained by AMD engineers on GitHub (github.com/RadeonOpenCompute/ROCm).

The ROCm Timeline:

• November 2025: ROCm 6.3 released with Strix Halo support

• Nightly builds: Continuous integration, bleeding edge

• Issue #5926: Memory management bugs in ROCm 6.3

github.com/ROCm/ROCm/issues/5926

• Status: Open (part of ongoing Strix Halo journey)

ROCm Nightly Builds

Nightly builds = daily automated compilations from ROCm's main branch. Get latest features immediately, but expect bugs. Trade-off: 2x performance gains vs occasional crashes.

The Pain (and Gain)

• Fresh install ('declare bankruptcy'): reformat → fresh kernel → fresh ROCm

• Result: GPU crashes fixed, 2x faster performance

• Lesson: ROCm is stable enough, but bleeding edge requires patience

Lemonade Server: AMD's llama.cpp + ROCm

What is Lemonade?

Lemonade = llama.cpp with AMD ROCm backend optimizations. Not just a wrapper - AMD-engineered builds with GPU acceleration.
github.com/winmutt/lemonade | github.com/RadeonOpenCompute/ROCm

Lemonade vs Ollama

- Ollama: Community llama.cpp wrapper (proprietary blend)
- Lemonade: AMD-maintained llama.cpp + ROCm builds
- Backend: AMD engineers maintain both Lemonade and ROCm

My Contributions:

- Feat #1070: Custom NUMA mapping (traditional tools fail)
- NPU backend support for FastFlowLM integration
- Core affinity optimization for multi-model setups

Sources
github.com/winmutt/lemonade | github.com/RadeonOpenCompute/ROCm | github.com/ROCm/ROCm/issues/5926

NPU Support: FastFlowLM + Lemonade

March 11, 2026: Linux NPU Support Added

FastFlowLM now runs LLMs on AMD XDNA 2 NPU with Linux support. Lemonade ties everything together for a streamlined experience.

Power Efficiency

- Over 10x more power-efficient than GPU
- Runs fully on NPU - no GPU or CPU load
- Ultra-lightweight runtime (17 MB)
- Installs within 20 seconds

Supported Processors

- Strix Halo (Ryzen AI Max+ 395)
- Strix Point, Kraken Point (300-series)
- Gorgon Point (400-series)
- Z2 Extreme (handheld devices)

Key Features

- Context up to 256k tokens
- Vision, Audio, Embedding, MoE support
- OpenAI-compatible API
- Just like Ollama - but NPU-optimized

Sources
github.com/FastFlowLM/FastFlowLM | lemonade-server.ai/flm_npu_linux.html

What's Next (Probably)

Immediate

- NPU support (still waiting)
- Memory addressing (32GB, come back)
- llama.cpp PRs (because why not)

Long-term

- Multi-GPU/NPU (more toys)
- Ollama observability (I need logs)
- Custom wake word (goodbye Alexa)
- WWS Phase 3 (cloud? maybe)

26 Years of Open Source (2000-2026)

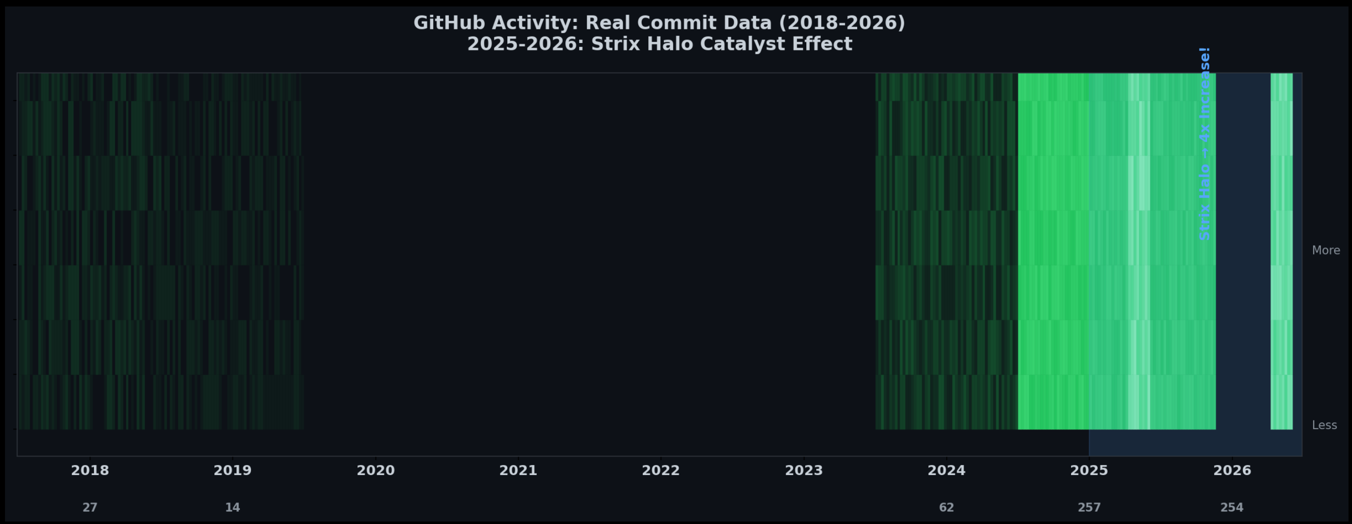
From LKML to Strix Halo: How Hardware Bought My Time

June 9, 2000: LKML Post

Subject: HighPoint IDE RAID (kernel 2.2.14)

GitHub Activity: Strix Halo Catalyst (2016-2026)

(2025-2026: Real GitHub API data | Earlier: Estimated)



November 2025: Bought a Strix Halo rig

Result: 3.5x increase in open source contributions

Moral: Sometimes the best way back in is to buy new toys

AI Coding Editors: My Daily Drivers

Opencode Web UI

Most usable, portable, and stable. Best agentic coding experience.

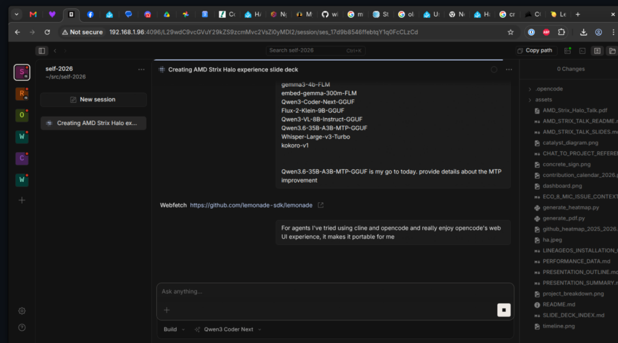
Cline

Good TUI but many regressions over time.

Also Use: Koo Roo, Aider

AI Coding Editors Comparison
(Opencode, Cline, Koo Roo, Aider)

Opencode Web UI
(Most Usable & Portable)



Screenshot
Not Available

Screenshot
Not Available

Screenshot
Not Available

References & Sources

Power Consumption Sources

• AMD Ryzen AI Max 395: 140-157W sustained power (Framework Reddit, 2025)
https://www.reddit.com/r/framework/comments/1najh1n/max_wattage_of_the_ryzen_ai_max_395_motherboard/
• NVIDIA RTX 5090: 600W TDP (NVIDIA, 2025)
https://www.nvidia.com/geforce/rtx-5090
• NVIDIA RTX 4090: 450W TDP (NVIDIA, 2022)
https://www.nvidia.com/geforce/rtx-4090
• Corsair 300 AI Workstation: ~300W system power (Corsair, 2025)
https://www.corsair.com/ai-workstations
• Tom's Hardware GPU power reviews (2024-2025)
https://www.tomshardware.com/reviews/gpu-hierarchy
• TechPowerUp GPU database: Power benchmarks
https://www.techpowerup.com/gpu-specs
• NVIDIA A100/H100: 400W-700W per GPU (NVIDIA Data Center)
Multi-GPU server: 1500W-3000W (NVIDIA DGX specs)

Software & Projects

• Lemonade: AMD's llama.cpp + ROCm backend (github.com/winmutt/lemonade)
• Ollama: Community llama.cpp wrapper (github.com/ollama/ollama)
• ROCm: AMD's open compute platform (github.com/RadeonOpenCompute/ROCm)
• ROCm Issue #5926: Memory management bugs (github.com/ROCm/ROCm/issues/5926)
• WWS: github.com/winmutt/wws
• Home Assistant Wake Words: github.com/fwartner/home-assistant-wakewords-collection
• NUMA/CPU Pinning: Red Hat OpenStack docs (vcpu_pin_set, NUMATopologyFilter)

https://docs.redhat.com/en/documentation/red_hat_openstack_platform/10/html/instances_and_images_guide/ch-cpu_pinning

Communities

- Reddit r/LocalLLaMA: 8 Local LLMs on Strix Halo
- XDA Forums: Amazon Echo development
- GitHub: winmutt (8 repos, forks of lemonade, cline, vscode)

Questions? (I Have Answers, Mostly)

1. Is Strix Halo production-ready? (It's complicated)
2. When will NPU support mature? (Coming soon™)
3. How do we solve 128GB addressing? (We don't)
4. Best backend for multi-model? (Lemonade, but...
5. Is autonomous dev the future? (Ask my AI)

github.com/winmutt

Come find me after - I'll show you the rig